

Chethan Reddy

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Email: achethanreddy1921@gmail.com

Mobile: +91-9030496717

EDUCATION

- **Vellore Institute of Technology**

Vellore, India

Bachelor of Engineering, Computer and Electrical Engineering

TECHNICAL EXPERIENCE

- **Staple AI — Forward Deployed Engineer (FDE-2)**

Singapore, Hybrid

Current role: September 2026 – Present

At Staple: May 2024 – Present

- **Solution architecture & delivery:** Own technical scoping and delivery for **20+ enterprise customers**; turn discovery findings into implementation plans with product engineering, customer success and QA.
- **Customer telemetry and AI:** Build AI dashboards and agents around customer telemetry to surface usage patterns and upsell opportunities; use these insights to support customer adoption and usage-growth discussions.
- **Customer enablement:** Delivered technical training and handover guides on extraction rules and data quality; translated customer requirements into API guidance and clarified responsibilities with implementation teams.
- **Production reliability:** Investigated ingestion and extraction failures using logs, document comparisons and root-cause analysis; led transactional repairs and monitored recovery with dry-run tooling and explicit verification checks.

Earlier: AI Software Engineer (SDE-2; previously SDE-1)

SDE-2 from May 2025

- **AI feature delivery:** As a core engineer, shipped complex-table extraction across **3 services and 3 regions**; built Python APIs, model/template schemas and per-page mapping of extracted cells to OCR boxes.
- **Distributed systems:** Co-designed and implemented async document processing with RabbitMQ, retries, deduplication and dead-letter handling; tested scan-message compatibility against **12 reference fixtures** in staging.
- **LLM evaluation:** Built an internal dashboard across **8 providers and 25 models** for ground-truth scoring (F1/similarity), repeat-run consistency and token-cost comparisons; added JSON validation and model routing.
- **QA automation:** Built regression tests for **70 API endpoints** with response contracts, negative cases and isolated-data cleanup; moved region-aware scheduling to GitHub Actions using Postman CLI.
- **Shared infrastructure:** Created a Python library for telemetry, database sessions and S3 access with **165 tests**; separated read/write traffic across **6+ services** with replicas, connection-pool tuning and distributed tracing.
- **Customer migrations:** Authored migration playbooks with baseline snapshots, extraction/export comparisons and rollback procedures; built dry-run tooling, idempotency checks and post-write verification for controlled releases.

- **Dotnitrone Technologies — AI Software Engineer**

USA, Remote

Alvarez & Marsal engagement: Diligence GPT

Dec 2023 – May 2024

- **RAG architecture:** Designed retrieval-augmented generation workflows for Diligence GPT with cross-functional teams, connecting document retrieval and AI-generated analysis for enterprise due-diligence use cases.
- **Backend leadership:** Led Python backend development with **Django, FastAPI and ArangoDB**, integrating AI tools and libraries into the application's services and database-backed retrieval workflows.
- **AI integration and quality:** Applied LangChain and OpenAI to backend features; reviewed code, debugged integrations and tested AI workflows. The Alvarez & Marsal client engagement continued through June 2024.

- **Bajaj Finserv Health — Data Science Engineer Intern**

Pune, India

InSightRx Data Team

May – Aug 2023

- **Recommendation systems:** Led development of user-friendly lab-test recommendations for the InSightRx data team, improving operational efficiency by **30%** in project measurements.
- **Healthcare image analytics:** Applied **LLMs, OCR, NER and YOLO** to healthcare documents and images, combining text recognition, named-entity extraction and object detection to support downstream data analysis.
- **Search optimization:** Refined Elasticsearch and SQL queries with fuzzy matching to improve access to healthcare information, achieving **80% faster retrieval** in project measurements.

- **TIFAC / VIT — Backend Development & Data Intern**

Vellore, India

Backend Lead, Physio Plus

Dec 2022 – Jun 2023

- **Backend ownership:** Led the complete Physio Plus backend using **FastAPI and MongoDB**; designed patient-record and clinic workflows for MicroClinics and supported analysis of patient data.
- **Scale and infrastructure:** Used AWS and a distributed architecture to support **100+ daily patient appointments**; also researched algorithms for autonomous drones in inventory management.

EARLIER EXPERIENCE AND MENTORING

- **Smart Diet Planner — Machine Learning Researcher** Remote
Research and development internship Aug 2022 – Apr 2023
 - **AI features:** Built search engines and automated PDF analysis with text and nutrition extraction for diet planning.
 - **Nutrition workflows:** Built macro-consumption calculations and nutrition analysis for personalized diet planning.
 - **Conversational AI:** Developed a customized **PaLM-based diet chatbot** for an application serving **850k+ users**, generating diet recommendations tailored to medical conditions and complications.
- **Grroom — Machine Learning Intern / Team Lead** India
Promoted to Team Lead in January 2022 Nov 2021 – Feb 2022
 - Trained YOLO for **100k+ iterations** for fashion-image detection and tagging, integrated models with the backend and led **10+ interns** developing ML algorithms and Python annotation scripts.
- **Tekie / Cybeorg / CipherSchools & other platforms** Remote
Python, ML and Software Engineering Mentor 2021 – 2023
 - **Teaching and code review:** Mentored **60+ students** through **500+ hours** at **Tekie** in Python/C++ and problem-solving; debugged coding submissions. At **CipherSchools**, reviewed Python projects and recommended improvements.
 - **Scientific programming:** Built **Cybeorg** curricula and delivered **300+ hours** in programming mathematics, ML, data science and astronomy; applied NumPy, SciPy and pandas in research-support projects and practical lessons.
 - **Software and ML mentoring:** Covered ML, databases, system design and algorithms at **edvi**; built ML curricula at **Modo Edulabs** and taught model development and Python/AI at **RoboGems (experience)** and **Camp K12**.

SELECTED ENGINEERING PROJECTS

- **eduAid / KaRmA / hAIr:** Built eduAid class analytics and note digitization (FastAPI/MongoDB), KaRmA's LangChain/SQL reports, and parsing, matching and interview-question features within the hAIr team.
- **Supplier Analytics / Insurance Search:** Analyzed **7m+ supplier records** using rating, cost and delivery data; built FastAPI search, autocomplete and recommendations with linear SVMs, Levenshtein distance and graph matching.
- **WeDio / FFCS Planner / PhotoBook:** Built **five-class CCTV disruption detection**, graph-based course timetables and image-to-PDF APIs with upload, resize and layout operations using FastAPI, Pillow and FPDF.

HACKATHON ACHIEVEMENTS

- **Winner, 1st Prize** — **Bitget U-30 Hackathon** (2023).
- **Winner, 1st Prize** — **IEEE Women in Engineering VIT WeHack 3.0, powered by Honeywell** (2022).
- **1st Runner-up** — **Axis Bankathon LLM 1.0** (2023).
- **1st Runner-up, 2nd Prize** — **HackRx 3.0, Bajaj Finserv Health** (2022).
- **Grand Finalist, Top 4 of 3,000+ teams** — **GLC-HP Hackathon** (2024).
- **Top 5 finalists** — **Canara Bank DACOE-Thon** (2024).
- **Grand Finalist, Top 7** — **SirionLabs HackFest** (2022); 2,000+ entries.
- **Top 7** — **Walmart Sparkathon** (2023).
- **Phase 2, Top 50 teams** — **Samsung Solve for Tomorrow, IIT Delhi** (2022); 18,000+ registrations.
- **Finalist** — **Siemens Healthineers SHIFT Hackathon** (2022).
- **Top 10 of 150+ teams** — **JSSATEB Hackwell 3.0, powered by Honeywell**.
- **Top 5** — **Amadeus H4S**; travel advisory using 3D graphs.
- **Top 10** — **Make-a-thon**; AI Sight for eye-condition screening.
- **Top 10** — **AI Arena 1.0**; data analytics with prompt engineering.
- **Top 100** — **SmartIdeathon** (2023); multimodal transportation algorithms.

TECHNICAL SKILLS

Architecture & leadership: Solution architecture, customer discovery/scoping, cross-functional technical leadership, mentoring, code review and enterprise API integration.

GenAI & LLMOps: RAG, prompt engineering, AI agents, LLM/VLM integration, structured outputs, model evaluation/routing and observability; embeddings/vector-search evaluation (Milvus), LangChain.

Software & platforms: Python, SQL, TypeScript/JavaScript, Go, C++; FastAPI, Django, Flask, React/Next.js; PostgreSQL, MongoDB, ArangoDB, Elasticsearch; RabbitMQ/Temporal; AWS, Kubernetes, OpenTelemetry, CI/CD.

AI & model training: TensorFlow/Keras, NumPy, SciPy, pandas; supervised learning, neural networks, model training and evaluation; YOLO object detection, computer vision, OCR, named-entity recognition, SVMs and regression.

Relevant credentials: [TensorFlow Hub NLP](#); [Computer Vision Basics](#); [AWS Machine Learning](#); [full credential record](#).

Spoken languages: English (fluent), Telugu and Hindi.